Hao Zhang

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LinkedIn: zhanghao27
Google Scholar: Profile

Education

The University of California, Los Angeles — Los Angeles, US
Postdoc in Electrical and Computer Engineering — 2026–Current

SLAC National Accelerator Laboratory & Stanford University — Menlo Park, US
Visiting Physicist at LCLS SRD Laser Science — 2022–Current

The University of California, Los Angeles — Los Angeles, US
PhD Researcher in Electrical and Computer Engineering, GPA: 3.93/4.0 — 2022–2026

The University of Victoria — Victoria, CA
MAS.c Student in Electric and Computer Engineering, GPA: 8.75/9 — 2020–2022

The University of British Columbia — Vancouver, CA
MAS.c Visiting Student (NSERC Quantum BC Scholar), GPA: 3.95/4.3 — 2020–2022

Tianjin University — Tianjin, China
BS.c in Electric and Computer Engineering (Integrated Circuit Design) — 2016–2020

Summary of Expertise

- **AI & Computational Physics:** AI for Science, Deep / Machine Learning, Physical AI, Reinforcement Learning, Multimodal AI, Agentic AI, Digital Twins, Optical Neural Networks
- **Nonlinear & Nanophotonics:** Nonlinear Optics, Nanophotonics, Metasurfaces, Plasmonics, Optical Trapping, Photonic Circuits, Quantum Optics & Computing
- **Ultrafast & Beam Physics:** Ultrafast Science, Accelerator Beam Physics, High-Field Light-Matter Interaction, high-brightness light sources

Selected Publications

1. **Zhang H**. Design, Prediction, and Optimization of Ultrafast Waveforms for Extreme Brightness Light Sources and Beyond. PhD thesis. UCLA, 2026
2. Hirschman J, Abedi E, Wang M, **Zhang H**, Borthakur A, Baker J, and Carbajo S. Deep learning-assisted modeling for $\chi^{(2)}$ nonlinear optics. Advanced Photonics 2026; 8:036004
3. **Zhang H**, Xu Y, Zhang W, Choudhary S, Alam MZ, Nguyen LD, Klein M, Vangala S, Miller JK, Johnson EG, Hendrickson JR, Boyd RW, and Carbajo S. Hybrid deep reconstruction for vignetting-free upconversion imaging through scattering in ENZ materials. Light: Science & Applications, 15, 327. 2026
4. Carbajo S, Bahk SW, Baker J, Bertozzi A, Borthakur A, Di Piazza A, Forbes A, Gessner S, Hirschman J, Lewenstein M, Li Y, Nam I, Otte E, Rosenzweig J, Shen Y, Song L, Tian Y, Wang Y, Wang Y, Wright L, Wu X, and **Zhang H**. Structured light at the extreme: harnessing spatiotemporal control for high-field laser-matter interactions. Advanced Photonics (in press); arXiv:2512.05042 2026

5. **Zhang H**, Lemons R, Hirschman J, Neveu NR, Robles R, Franz P, Mencer B, Sudar N, Britton M, Borne K, Zhang Z, Larsen KA, Baker J, Pennington C, Obaid R, Qiang J, Cryan J, Robinson J, Marinelli A, and Carbajo S. Upstream laser-based longitudinal enhancement of relativistic photoelectrons. Under review at Nature Photonics 2025
6. Hirschman J, Lemons R, **Zhang H**, Obaid R, Robles R, Franz P, Mencer B, Neveu NR, Britton M, Cesar D, Sudar N, Zhang Z, Baker J, Pennington C, Borne K, Driver T, Larsen KA, Guo V, Ding Y, Just G, Zhou F, Cryan J, Robinson J, Coffee R, Marinelli A, and Carbajo S. Advanced control of electron beams: tailoring X-ray production with programmable laser shaping. Under review at Nature Photonics 2026
7. **Zhang H**, Hirschman J, Lemons R, Neveu NR, Robinson J, Edelen AL, Raubenheimer TO, Wang D, Qiang J, and Carbajo S. Physics-guided inverse design of optical waveforms for nonlinear electromagnetic dynamics. Under review at Light: Science & Applications; arXiv:2603.12503 2026
8. **Zhang H**, Xu Y, Cui W, Boyd RW, and Carbajo S. Universal quantum interconnects via phase-coherent four-wave mixing. Advanced Photonics Nexus (in press) 2026
9. Xu Y, **Zhang H**, Zhang W, Niu L, Kulkarni G, Carbajo S, and Boyd RW. Physics-guided deep neural networks for full modal characterization of high-dimensional entangled states. Under review at Optica Quantum (co-first author) 2026
10. **Zhang H**, Mann J, Rosenzweig J, Chini M, and Carbajo S. Charged-particle control via spatio-temporally tailored pulses from gas-based nonlinear optics. Optical Materials Express 2026; 16:611–27
11. Zhang W, Tang Z, **Zhang H**, Rui S, Wang ZJ, and Wang X. Modality-Agnostic Federated Learning with Adaptive Updates for Heterogeneous Medical Image Tasks. IEEE Transactions on Medical Imaging 2026
12. **Zhang H**, Sun L, Hirschman J, and Carbajo S. Multi-modality deep learning for pulse prediction in homogeneous nonlinear systems via parametric conversion. APL Photonics 2025; 10
13. **Zhang H**, Zhao T, Zhang W, Halvaei S, Peters M, Cheung TS, Williams KC, and Gordon R. Accurate label-free classification of cancerous extracellular vesicles using nanoaperture optical tweezers and deep learning. npj Biosensing 2025; 2:33
14. Zhang W, **Zhang H**, Jiang Z, Servati A, and Servati P. Real-time forecasting of pathological gait via IMU navigation: a few-shot and generative learning framework for wearable devices. Discover Electronics 2025; 2:1–19
15. Sun L, **Zhang H**, Leary C, Lim C, Pennington C, Azcoitia G, Lu B, and Carbajo S. Antisymmetric chirp transfer to high-energy ultraviolet pulses via gas-based chirped four-wave mixing. Optics Letters 2025; 50:7460–3
16. **Zhang H**, Sun L, Hirschman J, Shariatdoust MS, Belli F, and Carbajo S. Optimizing spectral phase transfer in four-wave mixing with gas-filled capillaries. Optics Express 2024; 32:44397–412
17. Lemons R, Hirschman J, **Zhang H**, Durfee C, and Carbajo S. Nonlinear shaping in the picosecond gap. Ultrafast Science 2025; 4:0112
18. **Zhang H**, Gilevich S, Miahnahri A, Alverson SC, Brachmann A, Duris J, Franz P, Fry A, Hirschman J, Larsen KA, et al. The LCLS-II photoinjector laser infrastructure. High Power Laser Science and Engineering (Cover Feature & Editor's Pick) 2024 :1–33
19. **Zhang H**, Ma K, Zhang W, and Yan N. A novel self-packaged DBBPF with multiple transmission zeros for 5G sub-6 GHz applications. Microwave and Optical Technology Letters 2023; 65:62–8
20. **Zhang H**, Moazzezi P, Ren J, Henderson B, Cordoba C, Yeddu V, Blackburn AM, Saidaminov MI, Paci I, Hughes S, et al. Coupling perovskite quantum dot pairs in solution using a nanoplasmonic assembly. Nano Letters 2022; 22:5287–93

21. Zhang W, Ma K, **Zhang H**, and Fu H. Design of a compact SISL BPF with SEMCP for 5G sub-6 GHz bands. IEEE Microwave and Wireless Components Letters 2020; 30:1121–4

22. **Zhang H**. Solution-based analysis of individual perovskite quantum dots and coupled quantum dot dimers using nanoplasmonic tweezers. M.A.Sc. Thesis. University of Victoria, 2022

Selected Scholarships and Awards

International and National Recognitions

- ECE Dissertation Year Award 2026
- Frontiers in Optics + Laser Science (FiO+LS) Technical Group Poster Competition Award 2025
- US Particle Accelerator School Scholarship 2025
- IEEE Photonics Society Graduate Student Scholarship 2025
- SPIE Optics and Photonics Scholarship 2025
- Best Student Paper Award in CLEO-APS Division of Laser Science (DLS) competition (3rd place) 2025
- Best Student Paper Award in Optica Laser Congress and Exhibition / Advanced Solid State Lasers Conference (ASSL) 2024
- Optica Siegman School Participation Invite 2024
- SPIE Optics + Photonics Education Travel Scholarship 2023
- PSC Tingye Li Memorial Scholarship (Finalist) 2023
- Canada NSERC CREATE Quantum Computing Program Scholarship 2021 –2023
- China College Students Integrated Circuit Competition (North Region), Top 1 of 140 2019
- China College Students Integrated Circuit Competition (Final), Second Prize (Top 1%) 2019
- "Mathematical Contest in Modeling (MCM)" Honorable Mention 2017

University-Level Awards

- UCLA Doctoral Travel Grant 2025
- UCLA Henry and Susan Samueli Fellowship in Electrical Engineering 2023 –2024
- University of Victoria CAMTEC Breakthrough of the Month Award 2022
- University of Victoria Excellent Graduate Award 2021 –2022
- University of Victoria Graduate Student Fellowship 2020 –2023
- USRP Excellent Project Award 2019
- USRP Scholarship, Tianjin University 2018
- "Merit Student" Scholarship, Tianjin University 2017 –2018
- Science and Technology Talent Student Award, Tianjin University 2018

Leadership and Special Recognitions

- IEEE-UPP Best Student Branch Award in China (Co-chairman) 2018 –2019
- IEEE-UPP Best Student Branch Award in China (Co-chairman) 2017 –2018
- IEEE UBTECH Education Robotics Design Challenge (China), Silver Award 2017
- IEEE Future Elite of Science and Innovation Award, Tianjin University 2017

Research Experience

SLAC National Accelerator Laboratory — Menlo Park, US
Visiting Physicist at LCLS SRD| Supervisor: Dr. Joseph Robinson/Dr. Sergio Carbajo — Oct 2022 –Current

- Research on the development of the LCLS-II Photoinjector Laser Infrastructure (Sector 0) at LCLS Laser Science Group. The proposed upgraded photoinjector laser system has been fully installed at LCLS-II.
- Exploring UV shaping and resonant dispersive wave generation via gas-filled hollow-core fiber at **Dr. Tony Heinz's Lab**, *Stanford Pulse Institution.*
- Participate in developing the "Lightsource Unified Modeling Environment (LUME)," a start-to-end simulation platform designed to streamline the setup, execution, and analysis of experiments at advanced laser light sources;
- Participate in the beam-time as the qualified laser scientist:
  - [**L-10080/L-10137/L-10152/L-10322**] Phase-I/II/III/IV: Structural and Dynamic Basis for Quantum Effects in Enzyme Catalysis ;
  - [**L-10023**] Ultrafast Nonequilibrium Dynamics of Water under Strong-field Librational Excitation;

The University of California, Los Angeles — Los Angeles, US
PhD Student (LCLS subgroup) | Advisor: Prof. Sergio Carbajo — Jan 2023–Current

- Working on nonlinear optics and QEDs for X-ray laser applications with Machine Learning methods;
- Customize UV Pulses through Inverse Engineering Design Using Four-Wave Mixing in gas-filled Hollow-Core Fiber;
- Participate in the Federal funding program:
  - [**DE-AC02-76SF00515**] US Department of Energy, Free Electron Laser Research and Development; Stanford University & SLAC
  - [**DE-SC0022559**] US Department of Energy, DOE BES Detector & Accelerator, UCLA & SLAC
  - [**FA9550-23-1-0409**] US Department of Energy, BES Detector & Accelerator, UCLA & SLAC
  - [**N00014-24-1-2038**] US Office of Naval Research, UCLA & SLAC
  - [**NSF 2436343**] US National Science Foundation, Mathematical Foundations of AI-assisted Digital Twins for High Power Laser Science and Engineering, UCLA & SLAC
  - [**NSF 2431903**] US National Science Foundation, Nanoscale photonics technologies for ultra-compact X-ray source, UCLA

Nanoplasmonics Research Lab — Victoria, Canda
MAS.c student | Advisor: Prof. Reuven Gordon (FSPIE/FOPTICA/FIEEE) — Sep 2020 –Dec 2022

- Member of Centre for Advanced Materials and Related Technology (CAMTEC, UVIC)
- Studying nanoplasmonics and quantum optics under the guidance of **Prof. Reuven Gordon.** I worked on:
  - Second harmonic generation (SHG) from single perovskite crystal and epsilon-near-zero (ENZ) material;
  - Quantum tunneling effect for laser applications;
  - Exploring Dicke Superfluorescence from single Yb3+ nanoparticle;
  - Single/double quantum dots (CsPbBr3) trapping by using optical tweezers and their quantum effects.
- Selected by Canada NSERC QC-CREATE program (Organized by UBC, UVIC, and SFU) to learn quantum computing and information theories.
- Participate in the National funding program:
  - [**-**] NSERC (Canada) Discovery Grants program
  - [**-**] Canadian Foundation for Innovation (CFI)

Ansys.Inc - Lumerical Team — Vancouver, Canda
R&D Research Intern | Manager: Dylan McGuire — Jul 2021 –Dec 2021

- Research on developing an automated workflow for the extraction of compact models from physical simulation of MOSFETs/RF-SQUID that can be fabricated using CMOS-compatible processes. I was responsible for creating and validating simulation studies of the physical models that are necessary to characterize the MOSFET and SQUID using Ansys tools (HFSS, Nexxim, Lumerical) and building machine learning models for charge transport, and implementing electromagnetic modeling. **This achievement was committed and highly valued by ANSYS.Inc and was selected as the 2021 next generation of the development strategy.**

The University of British Columbia Vancouver, Canda
Visiting Student Sep 2020 –Dec 2022

- Research on Photonics Integrated Circuit Design;
- Research on Superconducting Integrated Circuit Design: the single qubit superconducting circuit (based on Transmon, fabricated with the Delta 1000 process of Star Cryoelectronics);
- Design a type of flexible antenna (dual-band) for personalized AI+Healthcare applications.

Tianjin University Tianjin, China
Research Student | Microwave-THz Laboratory (Advisor: Prof. Kaixue Ma (FIEEE/FCIE) - Dean)Mar 2018 –Jul 2020

- Research on designing self-packaged dual-band bandpass filter with multiple transmission zeros for 5G sub-6 GHz applications. (Excellent Undergraduate Thesis)
- Worked as a research assistant in the Heterogeneous IC research group of Tianjin Research Institute, China.

Conference and Presentations

- **Optical Shaping Meets Machine Learning** (**Invite talk**). 2026
  OPTICA Advanced Photonics Congress
- **Laser-Assisted Longitudinal Shaping of Relativistic Electron Beams** (**Oral**). 2026
  OPTICA Advanced Photonics Congress
- **Physics-Guided Optical Shaping for High-Brightness Photoemission** (**Poster**). 2026
  OPTICA Advanced Photonics Congress
- **Simultaneous characterization of two ultrashort pulses with near-zero-index material** (**Poster**). 2026
  Frontiers in Optics + Laser Science (FiO+LS)
- **Hybrid Deep Learning for Robust Upconversion Imaging within Epsilon-Near-Zero Environments** (**Poster**). 2026
  Frontiers in Optics + Laser Science (FiO+LS)
- **Adaptive Pump Modulation for OAM Entanglement via Machine Learning** (**Poster**). 2026
  Frontiers in Optics + Laser Science (FiO+LS)
- **Spectral phase quasi-linear transfer in gas-filled hollow-core fiber** (**Poster**). 2026
  Frontiers in Optics + Laser Science (FiO+LS)
- **Upstream Photocathode Laser Tailoring for Optimized Longitudinal Phase Space in High-Brightness FELs** (**Oral**). 2026
  33rd Linear Accelerator Conference (LINAC 2026)
- **Source-Level Laser Control for Tailored X-ray Pulses at LCLS-II** (**Poster**). 2026
  33rd Linear Accelerator Conference (LINAC 2026)
- **Enhanced Longitudinal Control of Relativistic Electrons via Spatiotemporal Laser Shaping** (**Oral**). 2026
  Conference on Lasers and Electro-Optics (CLEO)
- **Scattering-Resilient Upconversion Imaging Using Deep Hybrid Reconstruction in ENZ Media** (**Poster**). 2026
  Conference on Lasers and Electro-Optics (CLEO)
- **On-the-Fly Laser Shaping for Adaptive XFEL Operation** (**Oral**). 2026

Conference on Lasers and Electro-Optics (CLEO)

- **Phase transfer in dispersive four-wave mixing in gas-filled hollow capillary fibers** (**Talk**). 2026
  Conference on Lasers and Electro-Optics (CLEO)
- **Dynamic laser pulse shaping for future ultrafast electron and x-ray instrumentation** (**Oral**). 2026
  SPIE Photonics West (San Francisco, US)
- **Adaptive Laser Shaping for Next-Generation Ultrafast Electron and X-ray Instrumentation** (**Poster**). 2025
  Frontiers in Optics + Laser Science meeting, (Denver, USA)
- **Photoinjector Drive Laser System for LCLS-II** (**Oral**). 2025
  Frontiers in Optics + Laser Science meeting, (Denver, USA)
- **Leveraging the capabilities of LCLS-II: linking adaptable photoinjector laser shaping to tailored X-ray production** (**Oral**). 2025
  North America's premier particle accelerator conference, (Sacramento, USA)
- **Laser Shaping via Nonlinear Optics in Homogeneous Waveguides for High-Brightness Ultrashort Electron and X-ray Sources** (**Oral**). 2025
  Optica Nonlinear Optics Topical Meeting, (Honolulu, USA)
- **Simultaneous Nonlinear Frequency Conversion and Spectral Phase Transfer in Gas-Filled Fibers** (**Poster**). 2025
  Optica Nonlinear Optics Topical Meeting, (Honolulu, USA)
- **Indirect ultraviolet pulse shaping through upconversion in gas** (**Oral**). 2025
  Optica Nonlinear Optics Topical Meeting, (Honolulu, USA)
- **Advancing X-ray Sources with Digital Twin Surrogates for Ultrafast Optics** (**Poster**) 2025
  Ultrafast Optics XIV, (Portugal)
- **Linking Adaptable and Optimized Ultrafast Photoemission to Brighter X-rays** (**Oral**). 2025
  Ultrafast Optics XIV, (Portugal)
- **Ultraviolet pulse shaping through FWM-based line-symmetry upconversion** (**Poster**). 2025
  Ultrafast Optics XIV, (Portugal)
- **Deep Learning-Enhanced Study of Single Extracellular Vesicle Trapping Dynamics with Double-Nanohole Optical Tweezers** (**Invited talk**). 2025
  SPIE Optics+Photonics (San Diego, US)
- **Medical-informed Foundation Model for GBM Segmentation** (**Poster**). 2025
  SPIE Optics+Photonics (San Diego, US)
- **Laser Shaping for On-Demand High-Brightness Ultrashort Electron and X-ray Instrumentation** (**Poster**). 2025
  Conference on Lasers and Electro-Optics (CLEO)
- **The LCLS-II Photoinjector Laser System** (**Invite Talk**). 2025
  Conference on Lasers and Electro-Optics (CLEO)
- **Anti-symmetric dispersion control by frequency-mixing in gas-filled hollow capillary fibers** (**Oral**). 2025
  Conference on Lasers and Electro-Optics (CLEO)
- **Machine Learning Optimization of Nonlinear Dynamics in High-Power Laser Systems Using Digital Twins** (**Oral**). 2025
  Conference on Lasers and Electro-Optics (CLEO)
- **Simultaneous Narrow-Band Temporal Pulse Shaping and Up-Conversion in Gas-Filled Hollow-Core Fibers** (**Poster**). 2025
  APS Joint March and April Meeting Global Physics Summit
- **Closing the Picosecond Shaping Gap** (**Oral**). 2025
  APS Joint March and April Meeting Global Physics Summit

- **Overview of LCLS-II Photoinjector System** (**Oral**). 2025
  APS Joint March and April Meeting Global Physics Summit
- **Machine Learning for Digital Twin Surrogates of High-Power Laser Systems** (**Oral**). 2025
  APS Joint March and April Meeting Global Physics Summit
- **Asymmetric chirp control by four-wave-mixing in gas-filled hollow capillary fibers** (**Poster**). 2025
  APS Joint March and April Meeting Global Physics Summit
- **Prediction of ultrafast phenomena and nonlinear frequency conversion in hollow-core fibers Using deep learning** (**Poster**). 2025
  SPIE Photonics West (San Francisco, US)
- **Towards digital twins of high-power laser systems** (**Poster**). 2025
  SPIE Photonics West (San Francisco, US)
- **Chirp control by four-wave-mixing in gas-filled hollow capillary fibers** (**Poster**). 2025
  SPIE Photonics West (San Francisco, US)
- **Photoemission control with shaped UV pulses in free electron bunch generation** (**Poster**). 2025
  SPIE Photonics West (San Francisco, US)
- **The photoinjector laser system at LCLS-II** (**Oral**). 2024
  *Optica Advanced Solid State Lasers Conference (Osaka, Japan) - **Student Paper Finalist 1/14***
- **Simultaneous upconversion and spectral phase control in gas-filled fibers** (**Poster**). 2024
  *Optica Laser Applications Conference (Osaka, Japan) - **Best student paper award***
- **Enhanced deep learning approach for ultrafast nonlinear effects prediction in optical waveguides** (**Poster**). 2024
  The Siegman International School on Lasers (Menlo Park, US)
- **Enhanced photonic accelerator design for efficient Vector-Matrix multiplication in deep learning** (**Poster**). 2024
  SPIE Optics+ Photonics (San Diego, US)
- **Linking edge-ML X-ray diagnostics and adaptable photoinjector laser shaping for leveraging the capabilities of LCLS-II** (**Poster**). 2024
  15th International Particle Accelerator Conference (Barcelona, Spain)

Advancing attosecond X-ray sciences through four-wave mixing-driven photoemission (**Poster**).2024
Ultrafast Phenomena Conference, (Nashville, US)

- **A new integrated machine learning framework for advanced photoemission** (**Poster**). 2023
  SPIE Optics+ Photonics (San Diego, US)
- **Rapid analysis of single proteins and their interactions with accessible aperture-based tweezers** (**Invited talk**, on behalf of **Dr. Reuven Gordon**) 2022
  SPIE Optics+ Photonics Optical Trapping and Optical Micromanipulation XIX (San Diego, US)
- **Sizing single quantum dots insolvent using nano-tweezers** (**Poster**) 2022
  The 16th International Conference on Near-Field Optics (NFO-16), Nanophotonics and Related Techniques (Victoria, Canada)
- **Coupling between perovskite quantum dots in a plasmonic optical tweezer** (**Poster**) 2022
  The 16th International Conference on Near-Field Optics (NFO-16), Nanophotonics and Related Techniques (Victoria, Canada)
- **Comprehensive in-situ study of solution-based quantum dots with plasmonic tweezers** (**Poster**) 2022
  Nanophotonics and Micro/Nano Optics International Conference 2022, (Paris, France)

Teaching

- **Teaching Assistant** at University of Victoria — Winter 2022
  ECE 340 Applied Electromagnetics and Photonics (Prof. Tao Lu)
- **Teaching Assistant & Instructor** at Tianjin University — Summer 2018
  Microchip Control System (Prof. Jianrong Wang)
- **Teaching Assistant & Instructor** at Tianjin University — Summer 2017
  SWIFT: Programming in Apple products (Prof. Jianrong Wang)

Community Service

- **President and Co-Founder** — Nov 2017 –Sep 2019
  IEEE Student Branch at Tianjin University
  - Held the 14th and 15th Robot Competitions together with the Science and Technology Association;
- **Student Member** — 2017/2020–Current
  IEEE Photonics Society/SPIE/Optica/APS DLS/DPB
- **Session Chair (on behalf of Dr. Reuven Gordon)** — 2022
  SPIE Optics+ Photonics Optical Trapping and Optical Micromanipulation XIX, Session 4: Biophotonic Lab-on-a-Chip & Hybrid Systems III
- **Full Membership (nominated)** — 2025–Current
  The Scientific Research Honor Society (Sigma Xi)
- **Volunteer Reviewer** — Current
  Nature Communications, Optics Express, IEEE Microwave and Wireless Components Letters, Microwave and Optical Technology Letters, NeurIPS, AAAI, IJCNN, ISBI
- **Treasure** — 2020 –2022
  Optica and SPIE Student Chapter at University of Victoria
- **Teaching Assistant Lead** — Sep 2017 –Jun 2018
  Dawn Innovation Laboratory, Tianjin University
  - Training a group (10 new TAs) for teaching the 1st-year students how to use microcontroller systems such as Arduino and Raspberry PI 3/3B;
  - Demonstrating basic skills with Apple Products for teaching and learning. I was awarded the "Apple Teacher" by the Apple Vice President of Education.

Students Mentored

- **Ravi Varma** (MSc - Current: SLAC R&D)
- **Hongyi Yang** (Undergrad - Current: Amazon Engineer)
- **Tony Hong** (Undergrad - Current: UCLA)
- **Tiffany Chang** (Undergrad - Current: Amazon Engineer)
- **Alan Wang** (Undergrad - Current: CMU MS)
- **Jonathan Wang** (Undergrad - Current: UCLA)

References

Note: My confidential letters of recommendation are managed through Interfolio Dossier Delivery. To request a letter, please use the address labeled "Email for letter" below. If you have any questions for my letter writers, please feel free to contact them directly via their official institutional emails.

Referee	Title	Affiliation	Official email
Prof. Reuven Gordon	Professor	University of Victoria	rgordon@uvic.ca
Prof. Sergio Carbajo	Professor	University of California, Los Angeles	scarbajo@ucla.edu
Prof. Shukui Zhang	Lead Scientist	Jefferson Lab	shukui@jlab.org
Prof. Cuifeng Ying	Professor	Nottingham Trent University	cuifeng.ying@ntu.ac.uk
Dr. Ji Qiang	Lead Scientist	Lawrence Berkeley National Laboratory	jqiang@lbl.gov
Prof. Robert Boyd	Professor	University of Rochester	boydrw@me.com
Prof. Stephen Hughes	Professor	Queen's University	shughes@queensu.ca
Dr. Kirk Larsen	Staff Scientist	SLAC, Stanford University	larsenk@slac.stanford.edu
Dr. Nicole Neveu	Staff Scientist	SLAC, Stanford University	nneveu@slac.stanford.edu
Prof. Joseph Robinson	Staff Scientist	SLAC, Stanford University	jsrob@slac.stanford.edu
Dr. Dan Wang	Staff Scientist	Lawrence Berkeley National Laboratory	DWang2@lbl.gov
Prof. Kaixue Ma	Professor	Tianjin University	makaixue@tju.edu.cn
Prof. Makhsud Saidaminov	Professor	University of Victoria	msaidaminov@uvic.ca
Prof. Yu Luo	Professor	Tianjin University	yluo@tju.edu.cn